

Project 6: Market Prediction for Tesla

Problem Statement



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Overview and Summary

Please take a moment to picture yourself getting in your car in the morning, settling in to enjoy your coffee, and then being driven to work by it. Then you think of giving an old acquaintance a call or taking a look at the fantastic pictures from your most recent vacation. If you have to drive your car in heavy morning commuter traffic, it is not a good idea to do any of these things. However, we could use the time saved to do something more enjoyable or valuable if our automobiles were able to drive themselves. The emergence of self-driving cars is inevitable given the pace of technological advancement. Is this a favor or a sin? Road safety and improvements to the quality of life will ultimately win out, notwithstanding the necessity to address safety and policy problems throughout the development period.

Tesla was established in 2003 by a group of engineers who intended to show that driving an electric vehicle didn't require making sacrifices and that they could be superior to gasoline-powered automobiles in terms of performance, speed, and pleasure. Tesla now produces sustainable energy generating and storage devices that are infinitely scalable in addition to all-electric vehicles. Tesla contends that the sooner the world stops using fossil fuels and transitions to an emission-free future, the better.

The Roadster debuted Tesla's cutting-edge battery and electric drivetrain when it was introduced in 2008. From there, Tesla created the first luxury all-electric vehicle ever built from scratch.

Current State:

Since the demand for easy driving and the rising concern for security and safety lead to the need for high-end technology, culminating in the increased demand for autonomous and semi-autonomous vehicles, the autonomous vehicle is projected to dominate the US independent car market.



According to automation levels, the US market for autonomous vehicles is sizable at Levels 3 and 4. In Level 3 automation vehicles, the independent vehicle driving system performs all dynamic driving responsibilities without a driver. Active driving is part of braking, accelerating, keeping an eye on the vehicle, steering, and reacting to traffic-related incidents.

In Texas, Arizona, Washington, Michigan, California, and other states, self-driving cars have already been tested and used. However, they have limited movement.

Future State

In the United States, the current population is 320 million. Assuming the previous population growth rate, the number of people in 2025 is 340 million.

The commercial gig economy employs about 1% of the population in the US (Uber, Instacart, and so on). In the US, workers who directly support ground transportation for commercial shipping make up around 2% of the workforce.

According to the most recent Department of Labour figures, the median income in 2025 will be \$40,000. (Considering year-over-year inflation of 3-4 percent)

Tesla wants to grow and estimate the market size of driverless cars.

Sources:

<https://fred.stlouisfed.org/series/MEPAINUSA672N>

<https://data.worldbank.org/indicator/FP.CPI.TOTL.ZG?locations=US>

Task

Tesla is trying to estimate the market size of driverless cars. You are the PM at Tesla and have been given the responsibility to perform this task.

- Consider the United States as the primary geography for estimating



market size

- Assuming that the R&D and regulations will be ready in 2023, you need to estimate the market size for 2025.
- You should only consider consumer use-cases in your calculations and ignore commercial ones.
- You can use the internet to get the data points or numbers to calculate the final estimate.
- You can round off numbers during the estimation exercise for more straightforward calculations.